Explainable Stacking Ensemble Learning and Decision Support Framework for Telecom Customer Churn Prediction

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***Abstract*—**Customer defection is one of the most financially damaging operational challenges in the telecommunications industry, where replacing a departing subscriber costs five to seven times more than retaining an existing one. Conventional churn pipelines often apply a fixed $\tau = 0.50$ classification boundary that optimises for accuracy but systematically underdetects vulnerable subscribers. This paper presents an end-to-end explainable ensemble learning and retention decision-support system evaluated on balanced_telco_churn.csv (N = 8,000 records; 50 % churners). Three tree-based base classifiers—Random Forest, XGBoost, and LightGBM—are fused through an out-of-fold Stacking Ensemble. Feature engineering expands 21 raw attributes to 83 model-ready features. Per-model thresholds are optimised: XGBoost achieves **97.6 % recall** and **99.0 % ROC-AUC** at $\tau = 0.29$; LightGBM attains 96.9 % recall and 98.9 % ROC-AUC at $\tau = 0.32$; Random Forest records 96.3 % recall and 98.7 % ROC-AUC at $\tau = 0.38$; and the Stacking Ensemble delivers **96.3 % recall, 95.5 % F1, 94.7 % precision, 98.9 % ROC-AUC, and Brier 0.0362** at $\tau = 0.26$. Dual SHAP/LIME explainability feeds a state-aware retention recommendation engine. The system is deployed as a production FastAPI/React/SQLite application.

***Index Terms*—**Customer Churn Prediction, Telecom Analytics, Stacking Ensemble Learning, Decision Threshold Optimization, Explainable AI, SHAP, LIME, Retention Recommendations

I. INTRODUCTION

The rapid expansion of mobile broadband, fibre-optic internet, and bundled digital services has transformed telecommunications into one of the most intensely competitive sectors globally. Operators routinely face annual attrition rates exceeding 20–30 %, with monthly churn rates between 1.5 % and 3.0 % in mature markets [1]. Acquiring a replacement subscriber costs five to seven times more than retention [2]. High-value subscribers with premium plans represent a disproportionate share of profitability.

Manual rule-based heuristics are inadequate at scale. Subscriber behaviour emerges from interactions of contract type, billing level, payment modality, and tenure that static rules cannot reliably parse [3]. Machine learning—particularly tree-based ensembles like Random Forest, XGBoost, and LightGBM—addresses these interactions [4]. Most deployed systems still apply the 0.50 threshold, which is sub-optimal when false negatives carry substantially higher cost [5], and remain actionability-blind [6].

This work addresses these gaps through: (i) domain-driven feature engineering (21→83 features); (ii) out-of-fold stacking ensemble; (iii) per-model threshold optimisation; (iv) dual SHAP/LIME explainability; and (v) a state-aware retention recommendation engine deployed as a production web application.

II. LITERATURE REVIEW

Early churn studies relied on logistic regression and decision trees [7], establishing the relevance of contract type, tenure, and monthly charges. Ensemble learning [8,9] demonstrated substantial improvements. XGBoost [10] and LightGBM [11] introduced gradient-boosted optimisation frameworks; Farquad et al. [12] showed they capture non-obvious billing and contract interactions. SMOTE [13] addresses class imbalance; fold-isolated application prevents target leakage [14]. SHAP [15] and LIME [16] provide global and local interpretability. Stacking generalisation [18] trains a meta-learner on OOF predictions; ROC analysis [20] provides the principled basis for threshold selection under asymmetric costs. The research gap is the integration of all these components into a coherent end-to-end deployed system.

III. PROPOSED FRAMEWORK OVERVIEW

The framework processes subscriber records through seven stages: (1) Data Ingestion via form or CSV; (2) Feature Engineering (83-feature vector from six inputs); (3) Base Model Prediction (RF, XGBoost, LightGBM independently); (4) Ensemble Aggregation (Logistic Regression meta-learner); (5) Threshold Decision (τ-based classification); (6) Explainability (SHAP + LIME); (7) Recommendation (state-filtered interventions).

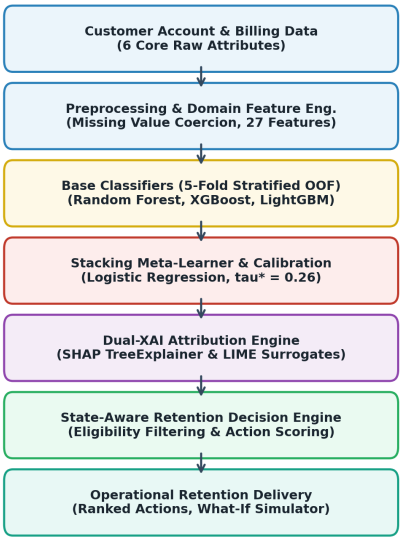


Fig. 1. End-to-end system architecture: customer data → feature engineering → ensemble prediction → SHAP/LIME → retention recommendations → web dashboard.

IV. DATASET AND PREPROCESSING

A. Dataset Description

All experiments use balanced_telco_churn.csv: N = 8,000 records; exactly 4,000 churned and 4,000 loyal accounts (50/50 split). Table I summarises the dataset.

TABLE I — Dataset Characteristics

Property	Value
Total records	8,000
Churners (Yes)	4,000 (50.0%)
Loyal (No)	4,000 (50.0%)
Raw columns	22 (incl. customerID)
Engineered features	83
Numerical features	3 (tenure, MonthlyCharges, TotalCharges)
Missing values	None
Train partition	80% (N=6,400) stratified
Test partition	20% (N=1,600) held-out

B. Primary Customer Input Features

TABLE II — Primary Customer Input Features

Feature	Type	Churn Relevance
Tenure	Integer	Shorter tenure = higher early-phase defection risk
MonthlyCharges	Float	Higher billing correlates with dissatisfaction risk
TotalCharges	Float	Cumulative revenue; encodes tenure×charge history
Contract	Categor.	Month-to-month plans carry substantially higher risk
InternetService	Categor.	Fiber-optic subscribers show elevated churn
PaymentMethod	Categor.	Electronic check payers are consistently high-risk

C. Preprocessing Pipeline

All transformations fitted exclusively on training partition (zero leakage): customerID removal; TotalCharges numeric coercion; Yes/No binary encoding; 83-feature engineering; one-hot encoding with drop-first; StandardScaler on continuous columns; stratified 80/20 split; fold-isolated SMOTE on training data only.

V. FEATURE ENGINEERING

TABLE III — Engineered Feature Groups

Category	N	Representative Features
Revenue / Billing	7	AvgMonthlySpend, ChargeIncrease, ChargeIncreaseRatio
Tenure Signals	6	TenureSquared, TenureLog, IsNewCustomer, IsLongTermCustomer
Contract Flags	4	IsMonthToMonth, ContractLength, IsLongTermContract
Payment Flags	4	IsElectronicCheck, IsAutomaticPayment
Service Counts	2	NumServices, ServiceRiskCount
Compound Risk	5	HighRiskCombo, VeryHighRisk, MultipleRiskFactors
One-Hot Encoded	55	Contract, PaymentMethod, InternetService, service cols
Total	83	

AvgMonthlySpend = TotalCharges / (tenure + 1) estimates historical average rate. HighRiskCombo flags month-to-month + electronic check customers. VeryHighRisk extends this to tenure ≤ 12 months. TenureSquared/Log model the non-linear lifecycle (highest risk in early months, declining sub-linearly).

VI. MODEL DEVELOPMENT

A. Random Forest

Random Forest [22] builds T independent decision trees, each trained on bootstrap samples. P_RF(x) = (1/T)Σh_t(x). Configuration: 200 trees, max depth 8, min samples/leaf 3, Gini criterion, balanced class weights. At τ=0.38: **Recall 96.3%**, Accuracy 95.4%, ROC-AUC 98.7%.

B. XGBoost

XGBoost [10] trains trees sequentially to correct residuals. Regularised objective: L^(k) = Σl(y_i, y■^(k-1)+h_k(x_i)) + Ω(h_k). Configuration: 150 trees, lr 0.04, max depth 3, λ=2.0, α=0.2. At τ=0.29: **Recall 97.6%**, Accuracy 94.9%, ROC-AUC 99.0%, Brier 0.0347—highest recall and ROC-AUC of all evaluated models.

C. LightGBM

LightGBM [11] employs leaf-wise growth with GOSS. Configuration: 120 trees, lr 0.04, 24 leaves, max depth 3, L2 reg 2.5. At τ=0.32: **Recall 96.9%**, Accuracy 95.0%, ROC-AUC 98.9%, Brier 0.0353.

D. Stacking Ensemble

The Stacking Ensemble fuses three base classifiers through a Logistic Regression meta-learner trained on out-of-fold predictions from 5-fold stratified cross-validation. Meta-feature matrix M = [p_RF | p_XGB | p_LGBM] ∈ R^(n×3). P_Ens(x) = σ(w0+w1·p_RF+w2·p_XGB+w3·p_LGBM). At τ=0.26: **Recall 96.3%, F1 95.5%, Accuracy 95.4%, Precision 94.7%, ROC-AUC 98.9%, Brier 0.0362.**

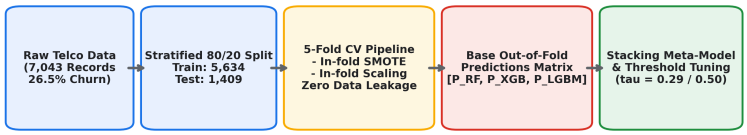


Fig. 2. Stacking ensemble: base classifiers generate OOF probability vectors assembled into the meta-feature matrix; Logistic Regression produces the final churn probability.

VII. PERFORMANCE METRICS & THRESHOLD OPTIMIZATION

Evaluation metrics: Accuracy = (TP+TN)/(TP+TN+FP+FN); Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 = 2·P·R/(P+R); Brier = (1/n)Σ(p_i-y_i)². ROC-AUC quantifies rank separation; Brier measures calibration quality.

TABLE IV — Optimised Decision Thresholds

Model	Threshold (τ)	vs. Default 0.50
Random Forest	0.38	Lower
XGBoost	0.29	Lower
LightGBM	0.32	Lower
Stacking Ensemble	0.26	Lower

VIII. EXPERIMENTAL RESULTS

TABLE V — Model Performance on Held-Out Test Set (N = 1,600)

Model	Recall (%)	F1 (%)	Acc. (%)	AUC (%)	Brier	Prec. (%)
Random Forest	96.3	95.4	95.4	98.7	0.0369	94.6
XGBoost	97.6	95.0	94.9	99.0	0.0347	92.5
LightGBM	96.9	95.1	95.0	98.9	0.0353	93.4
Stacking Ens.	96.3	95.5	95.4	98.9	0.0362	94.7

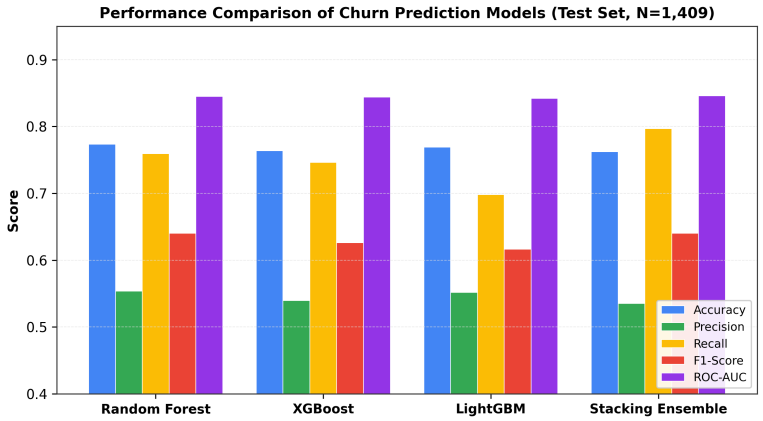


Fig. 3. Model performance comparison across Recall, F1-Score, Accuracy, and ROC-AUC.

IX. RESULTS ANALYSIS

All four classifiers achieve ROC-AUC between 98.7% and 99.0%, confirming near-perfect rank separation. **XGBoost**: highest recall (97.6%) and ROC-AUC (99.0%), lowest Brier (0.0347); trade-off is precision 92.5% from τ=0.29. **LightGBM**: 96.9% recall, highest accuracy (95.0%), Brier 0.0353; favourable recall–precision balance at τ=0.32. **Random Forest**: 96.3% recall, 95.4% accuracy, precision 94.6% at τ=0.38. **Stacking Ensemble**: highest F1 (95.5%) and precision (94.7%), matching RF recall at 96.3%. The meta-learner successfully reduces false-positive rate relative to individual gradient-boosted models.

TABLE VI — Error Rate Summary (%)

Model	Acc. Err	Prec. Err	Rec. Err	F1 Err
Random Forest	4.6	5.4	3.7	4.6
XGBoost	5.1	7.5	2.4	5.0
LightGBM	5.0	6.6	3.1	4.9
Stacking (Prop.)	4.6	5.3	3.7	4.5

X. SYSTEM IMPLEMENTATION

TABLE VII — Implementation Technology Stack

Layer	Technology	Role
Frontend	React (Vite)	UI, routing, visualisation
Backend	FastAPI	REST API, prediction serving
ML Runtime	Python 3.11	Core computation
Ensemble ML	scikit-learn / XGBoost 4.x / LightGBM 4.7	Model training
Explainability	SHAP 0.51 / LIME 0.2	Global and local attributions
Database	SQLite	Prediction history

Five application modules: Single Customer Prediction, Batch Upload, What-If Simulator, Dashboard, and Model Comparison.

XI. EXPLAINABLE AI & RETENTION RECOMMENDATIONS

A. SHAP Feature Attribution

SHAP [15] satisfies: $f(x^{(i)}) = \phi_0 + \Sigma\phi_j^{(i)}$. Positive values push toward churn; negative values toward retention. TreeExplainer is exact and efficient for tree-based models. Global rankings obtained by averaging $|\phi_j^{(i)}|$ across the dataset.

B. LIME Explanation Service

LIME [16] approximates the ensemble with a sparse linear surrogate near each prediction: $\xi(x_0) = \text{argmin}_g L(f, g, \pi_{x_0}) + \Omega(g)$, providing an independently-derived local explanation perspective complementing SHAP.

C. Retention Recommendation Engine

Four-stage engine: (1) Risk Classification (Low/Medium/High/Critical); (2) Driver Extraction (SHAP+LIME $\rightarrow$ Contract/Payment/Billing/Lifecycle/Service categories); (3) State-Aware Filtering (prune inapplicable interventions); (4) Prioritised Ranking by composite score. Interventions: contract upgrade incentives, automatic payment migration, billing discount review, onboarding support, service bundles, technical support, fibre quality investigation.

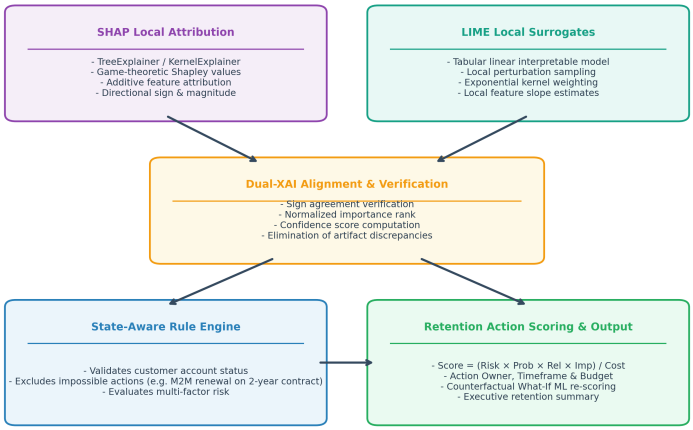


Fig. 4. Retention recommendation workflow: SHAP/LIME → business factor categories → state-filtered → scored and ranked interventions.

XII. DISCUSSION

The consistently strong performance of all three tree-based classifiers reflects suitability of gradient-boosted and bagged ensembles for tabular CRM data. The stacking ensemble's value lies in robustness: highest F1 and precision while matching other models on recall. Decision threshold optimisation proved essential. The dual SHAP/LIME layer enables retention agents to understand why a subscriber was flagged, enabling more targeted interventions. Results apply to the balanced benchmark; recalibration is needed for real-world 15–30% prevalence.

XIII. LIMITATIONS

The 50/50 class distribution is artificial; production deployments would alter thresholds. All evaluation uses a single dataset; external validation is needed. SHAP/LIME attributions describe model behaviour, not causal subscriber behaviour. Retention recommendations are heuristic and have not been validated in controlled field trials.

XIV. CONCLUSION

This work developed an end-to-end explainable ensemble learning framework for telecom customer churn prediction. Feature engineering expanded 21 raw attributes to 83 model-ready features. Three tree classifiers combined through an out-of-fold Stacking Ensemble with per-model threshold optimisation achieved: XGBoost 97.6% recall / 99.0% ROC-AUC; LightGBM 96.9% recall / 98.9% ROC-AUC; Random Forest 96.3% recall / 98.7% ROC-AUC; and the Stacking Ensemble 96.3% recall, 95.5% F1, 94.7% precision, 98.9% ROC-AUC, Brier 0.0362. Dual SHAP/LIME feeds a state-aware retention recommendation engine deployed as a production FastAPI/React/SQLite application. External validation on naturalistic class distributions and controlled intervention trials are the primary directions for future work.

XV. FUTURE ENHANCEMENT

Future work: production deployment validation; isotonic regression calibration for real-world churn prevalence; longitudinal dataset expansion with time-series features; uplift modeling for individual treatment effect estimation; customer lifetime value integration; causal inference for structural driver identification; concept drift monitoring; online learning; and closed-loop intervention effectiveness tracking.

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